Daffodil International University
Faculty of Engineering
Department of Information and Communication Engineering (ICE)
B.Sc. in ICE Program

COURSE OUTLINE- Optical Fiber Communication

PART A - Introduction

1. Course Code : ICE 3347
2. Course Title : Optical Fiber Communication
3. Course Type : Core Course
4. Level/Term and Section : L3T3
5. Academic Session : 2025-2026 and onwards
6. Course Instructor : Md. Sakaid Hossain Shakir (ID: 710003867), Initial: SHS, Department of ICE.
7. Prerequisite : N/A
8. Credit Value : 3.0
9. Contact Hours : 3 hours per week
10. Total Marks : 100
11. Course Objectives and Summary:

Optical fiber communication is the backbone of modern telecommunication, Internet and data-center networks. This course introduces the physical principles of light guidance in optical fibers and builds up to the analysis and design of complete fiber-optic links. Topics include the fundamentals of light (reflection, refraction, total internal reflection and numerical aperture), fiber structure and types, ray, wave and mode theory of propagation, attenuation and dispersion mechanisms, optical sources (LED and laser diode), optical detectors (PIN and APD), optical transmitters and receivers, splicing, connectors and coupling, fiber-optic measurements including OTDR, and system design using power and rise-time budgets. The course concludes with wavelength division multiplexing (WDM), optical amplifiers and optical fiber networks, giving students a thorough grounding in the technologies currently used to build high-capacity optical communication systems.

Objective:

The objectives of this course are

- To develop a solid understanding of the basic concepts of optical fiber communication, including light propagation mechanisms and fundamental system principles.
- To analyze different types of optical fibers, propagation modes, and the loss and dispersion mechanisms that limit the performance of fiber-optic links.
- To study the operating principles and characteristics of optical sources, detectors, transmitters, receivers, amplifiers and passive components.
- To develop the ability to design optical fiber communication systems and networks that meet given performance requirements using power budget and rise-time budget analysis.

12. Course Learning Outcomes (COs/CLOs):

At the end of the course, a student should be able to -

CLO 1	**Explain the basic concepts of optical fiber communication, including light propagation mechanisms and fundamental system principles.**
CLO 2	**Analyze different types of optical fibers, modes, optical sources, and detectors to evaluate their characteristics and performance in communication systems.**
CLO 3	**Design optical fiber communication systems by selecting appropriate fibers, sources, detectors, amplifiers, and network components to meet given performance requirements.**

13. Mapping of Course Learning Outcomes (COs/CLOs) with Program Outcomes (POs)
CLO/CO-PLO/PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CLO 1	3											
CLO 2		3										
CLO 3			3		2							

14. Teaching Learning Strategies (TLSs)

TLS1	**Interactive Lecture/discussion using Online/multimedia or whiteboard.**
TLS2	**Group presentation regarding related problems and assigned tasks.**
TLS3	**Evaluation of class performances of each student in a class for the topic.**

PART-B: Course Content

SL. NO.	Course Contents (as Summary)	Time Frame	TLSs	CLOs	Assessment Techniques
1	**Introduction to Optical Fiber Communication** Evolution of optical communication, Advantages, Disadvantages and Applications, Basic optical communication system and its components	**Week 1**	**TLS 1 TLS 2 TLS 3**	**CLO 1**	**Quiz, Assignment**
2	**Fundamentals of Light** Reflection and Refraction, Snell's law, Critical angle, Total Internal Reflection (TIR)	**Week 2**	**TLS 1 TLS 2 TLS 3**	**CLO 1**	**Quiz, Assignment**
3	**Acceptance angle and Numerical Aperture (NA); Optical Fiber Structure** Core, Cladding and Coating, Principle of light propagation, Meridional and Skew rays	**Week 3**	**TLS 1 TLS 2 TLS 3**	**CLO 1**	**Quiz, Assignment**
4	**Optical Fiber Types** Modes of propagation, Single-mode and Multimode fibers, Step-index and Graded-index fibers, Fiber classifications	**Week 4**	**TLS 1 TLS 2 TLS 3**	**CLO 2**	**Quiz, Assignment**

SL. NO.	Course Contents (as Summary)	Time Frame	TLSs	CLOs	Assessment Techniques
Quiz # 1: Based on Week 1-Week 4 Lessons. Achieve CLO1 and CLO2					
5	**Propagation of Light in Fiber** Ray theory, Wave theory and Mode theory, V-number (normalized frequency), Number of modes	**Week 5**	**TLS 1 TLS 2 TLS 3**	**CLO 2**	**Quiz, Assignment**
6	**Single-mode Fiber Parameters** Mode field diameter, Cutoff wavelength, Single-mode condition, Modal propagation and Group velocity	**Week 6**	**TLS 1 TLS 2 TLS 3**	**CLO 2**	**Quiz, Assignment**
7	**Fiber Losses/Attenuation** Intrinsic and Extrinsic absorption, Rayleigh and Mie scattering, Macrobending and Microbending losses, Attenuation coefficient, Loss measurement	**Week 7**	**TLS 1 TLS 2 TLS 3**	**CLO 2**	**Quiz, Assignment**
Quiz # 2: Based on Week 5-Week 7 Lessons. Achieve CLO1 and CLO2					
8	**Dispersion** Modal, Material, Waveguide and Chromatic dispersion, Pulse broadening, Bandwidth-distance relationship	**Week 8**	**TLS 1 TLS 2 TLS 3**	**CLO 2**	**Quiz, Assignment**
Mid Term: Based on Week 1-Week 8 Lessons. Achieve CLO1 and CLO2					
9	**Optical Sources** LED and Laser diode, Operating principles, Characteristics, LED vs. Laser	**Week 9**	**TLS 1 TLS 2 TLS 3**	**CLO 2**	**Quiz, Assignment**
10	**Optical Detectors** PIN photodiode, Avalanche photodiode (APD), Responsivity, Quantum efficiency, Detector noise, PIN vs. APD	**Week 10**	**TLS 1 TLS 2 TLS 3**	**CLO 2**	**Quiz, Assignment**
Quiz # 3: Based on Week 9-Week 10 Lessons. Achieve CLO2					
11	**Optical Transmitter and Receiver** Basic transmitter architecture, Modulation, Receiver architecture, Receiver sensitivity, Noise and BER	**Week 11**	**TLS 1 TLS 2 TLS 3**	**CLO 3**	**Quiz, Assignment Presentation**
12	**Fiber Splicing, Connectors and Coupling** Fusion splicing, Mechanical splicing, Connectors, Coupling losses, Optical couplers	**Week 12**	**TLS 1 TLS 2 TLS 3**	**CLO 3**	**Quiz, Assignment Presentation**
Quiz # 4: Based on Week 11-Week 12 Lessons. Achieve CLO2 and CLO3					

SL. NO.	Course Contents (as Summary)	Time Frame	TLSs	CLOs	Assessment Techniques
13	**Optical Communication System Design** Power budget, Rise-time budget, Link loss, System margin, Transmission distance	**Week 13**	**TLS 1** **TLS 2** **TLS 3**	**CLO 3**	**Quiz, Assignment Presentation**
14	**Fiber-Optic Measurements** Optical power measurement, Loss measurement, Optical Time Domain Reflectometer (OTDR), Fault location measurement	**Week 14**	**TLS 1** **TLS 2** **TLS 3**	**CLO 3**	**Quiz, Assignment Presentation**
15	**WDM, Optical Amplifiers and Fiber Networks** Wavelength Division Multiplexing (WDM), Optical amplifiers (EDFA, SOA, Raman), Optical networks (SONET/SDH, PON and FTTx)	**Week 15**	**TLS 1** **TLS 2** **TLS 3**	**CLO 3**	**Quiz, Assignment Presentation**
Final Exam: Based on Week 9-Week 15 Lessons. **Achieve CLO2 and CLO3**					

PART-C: Assessment and Evaluation

15. Assessment Strategy

Quiz:
Altogether 4 quizzes may be taken during the semester, 2 quizzes will be taken for midterm and 2 quizzes will be taken for final term. Best of 3 quizzes will be considered and marks will be average. No makeup quizzes will be taken. Students are strongly recommended not to miss any quizzes.

Assignment:
The students will have to submit the assignment individually. The topic, numerical problems or link-design case studies will be given as assignments during the class which they have to prepare at home and will submit on or before the due date. No late submission of assignments will be accepted. Students will have to do the presentation on the given topic as an assignment.

16. Assessment Procedure:

CIE (Continuous Internal Evaluation)- 60 Marks

Bloom's Criteria	Attendance (07)	Quiz (15)	Assignment (5)	Presentation (8)	Mid Term Exam (25)
Remember		**5**			**5**
Understand		**5**		**2**	**5**
Apply		**5**	**2**	**2**	**10**
Analyze				**2**	**5**
Evaluate				**2**	
Create			**3**		

SEE (Semester End Examination)-40 Marks

Bloom's Criteria	Marks (40)
Remember	5
Understand	10
Apply	10
Analyze	10
Evaluate	
Create	5

17. Evaluation Policy

Grades will be calculated as per the university grading structure and individual students will be evaluated based on the following criteria with respective weights.

Assessment Task	Marks (Total = 100)
Attendance	7
Class Test	15
Assignment	5
Presentation	8

Mid term Exam	25
Final Exam	40
Total	**100**

18. Make-up Procedure

In the case of unavoidable situations, make-up class will be conducted within 2 (two) weeks from the day of class via online or offline.

PART-D: Learning Resources

19. Textbook

- Gerd Keiser, Optical Fiber Communications, 5th Edition, McGraw Hill Education.

20. Reference Books and Materials

- John M. Senior, Optical Fiber Communications: Principles and Practice, 3rd Edition, Pearson Education.
- Govind P. Agrawal, Fiber-Optic Communication Systems, 4th Edition, Wiley.
- Djafar K. Mynbaev and Lowell L. Scheiner, Fiber-Optic Communications Technology, Pearson Education.

21. Other Resources

- Blended Learning Center (BLC): https://elearn.daffodilvarsity.edu.bd/